Hw 7

1.

(a)

poset A: the diagram is the chain $c \prec b \prec a$ so this means

$$R_A = \{(a,a),(b,b),(c,c),(c,b),(b,a),(c,a)\}$$

poset B this is $(\mathcal{P}(\{a,b\}), \subseteq)$ on the four elements $\emptyset, \{a\}, \{b\}, \{a,b\}$ so then

$$\begin{aligned} R_B = \{&(\emptyset,\emptyset),(\{a\},\{a\}),(\{b\},\{b\}),(\{a,b\},\{a,b\}),\\ &(\emptyset,\{a\}),(\emptyset,\{b\}),(\emptyset,\{a,b\}),\\ &(\{a\},\{a,b\}),(\{b\},\{a,b\})\} \end{aligned}$$

poset C: the cover relations are $1 \prec 3$, $1 \prec 4$, and $2 \prec 4$, so

$$R_C = \{(1,1),(2,2),(3,3),(4,4),(1,3),(1,4),(2,4)\}$$

(b)

poset A:

- Minimal elements: $\{c\}$
- Maximal elements: $\{a\}$
- Least element: c
- Greatest element: a

Poset (B):

- Minimal elements: $\{\emptyset\}$
- Maximal elements: $\{\{a,b\}\}$
- Least element: $\emptyset$
- Greatest element: $\{a,b\}$

poset (C):

- Minimal elements: $\{1,2\}$
- Maximal elements: $\{3,4\}$
- Least element: does not exist
- Greatest element: does not exist

(c)

1. for 1 and 2:
$$\mathrm{GLB}(1,2) \text{ does not exist}, \qquad \mathrm{LUB}(1,2)=4.$$
2. for 3 and 4:
$$\mathrm{GLB}(3,4)=1, \qquad \mathrm{LUB}(3,4) \text{ does not exist}.$$
3. for 1 and 4:
$$\mathrm{GLB}(1,4)=1, \qquad \mathrm{LUB}(1,4)=4.$$

2.

- **(A)** is a lattice
- **(B)** is a lattice
- **(C)** is not a lattice and for the pair $5,6$ theres no GLB because there is no common lower bound of 5 and 6
- **(D)** is a lattice
- **(E)** isnt a lattice and for the pair $4,5$ there is no LUB because the common upper bounds are $2,3,1$ and the least one does not exist since 2 and 3 are incomparable minimal upper bounds
- **(F)** is a lattice
- **(G)** isnt a lattice and for the pair $4,5$ there is no LUB because the common upper bounds are $2,3,1$ and the least one doesnt exist since 2 and 3 are incomparable minimal upper bounds

3.

(a) $(S,\preceq)$ **where** $S=\{1,2,3,5,6,10,15,30\}$ **and** $x\preceq y \iff x\mid y$

i. the cover relations are
$$1\prec 2,\ 1\prec 3,\ 1\prec 5,$$
$$2\prec 6,\ 2\prec 10,\ 3\prec 6,\ 3\prec 15,\ 5\prec 10,\ 5\prec 15,$$
$$6\prec 30,\ 10\prec 30,\ 15\prec 30.$$
so the Hasse diagram has 1 at the bottom and $2,3,5$ above it then $6,10,15$ and 30 at the top.

ii. yes,for any two elements of S both their greatest common divisor and least common multiple are also in S so every pair has both a meet and a join

iii. in this poset we got
$$x\wedge y=\gcd(x,y), \qquad x\vee y=\mathrm{lcm}(x,y).$$
eexamples:
$$6\wedge 10=2, \qquad 6\vee 10=30.$$
$$10\wedge 15=5, \qquad 10\vee 15=30.$$

(b) $(S, \preceq)$ **where** $S = \mathcal{P}(\{1,2,3\})$ **and** $x \preceq y \iff x \subseteq y$

i. the elements are

$$\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}.$$

the hasse diagram is the boolean lattice B_3: $\emptyset$ at the bottom and the three 1-element subsets above it so the three 2element subsets above those and $\{1,2,3\}$ at the top

ii. yes it's a latisse because every pair of subsets has both an intersection and a union so every pair has both a meet and a join

iii in this poset we have

$$A \wedge B = A \cap B, \qquad A \vee B = A \cup B.$$

examples:

$$\{1,2\} \wedge \{2,3\} = \{2\}, \qquad \{1,2\} \vee \{2,3\} = \{1,2,3\}.$$
$$\{1\} \wedge \{1,3\} = \{1\}, \qquad \{1\} \vee \{1,3\} = \{1,3\}.$$